Classical training of conditional instantaneous quantum polynomial-time Born machines

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Instantaneous quantum polynomial-time (IQP) Born machines admit classical estimation of training observables for suitable input states, allowing model optimization without quantum circuit evaluations. However, their standard fixed-input formulation models a single distribution. In this work, we introduce a conditional framework that encodes conditions into input states before applying a learned IQP unitary shared across conditions. The input preparation need not have strict IQP form, provided its reference state supports efficient classical sampling and amplitude-ratio evaluation. These operations enable unbiased classical estimation of Pauli-Z moments and their gradients. Experiments on a sixteen-qubit discrete Gaussian family and the Curie–Weiss model demonstrate conditional generation, with residual errors over the full distribution. Classical training remains computationally feasible in runs from 64 to 2048 qubits. Furthermore, we show how input-state encoding can impose exact output symmetries in both conditional and unconditional IQP Born machines.

I. INTRODUCTION

Conditional generative models approximate a family of target distributions $p(\cdot \mid c)$ with a model $q_{\boldsymbol{\theta}}(\cdot \mid c)$ whose learned parameters are shared across conditions. Classical examples include conditional generative adversarial networks [1–3], conditional variational autoencoders [4–6] and conditional diffusion models [7–9]. In the quantum domain, conditional variants of quantum circuit Born machines (QCBMs) [10, 11] address such tasks [12]. Conditional QCBMs encode conditions through input-state preparation or dedicated registers before a shared parameterized circuit [13–16]. Postselection on a learned joint Born distribution also yields conditional samples [17]. Beyond QCBMs, quantum diffusion models have explored conditional generation of both classical data and quantum states [18–20].

Sampling from certain output families of instantaneous quantum polynomial-time (IQP) Born machines is conjectured to be classically intractable under standard complexity-theoretic assumptions [21–25]. Yet, for training via the maximum mean discrepancy (MMD) objective [26], which is common for QCBMs [10, 27], the required Pauli-Z expectation values and parameter gradients admit efficient classical estimators [28]. These estimators build on Van den Nest's techniques for computationally tractable states [29]. However, the standard formulation models a single target distribution per circuit. Extending IQP Born machines to conditional generation requires a conditioning mechanism that preserves classical access to the training observables.

In this work, we develop a classically trainable framework for conditional IQP Born machines. A learned unitary $U_{\boldsymbol{\theta}}$ acts on the condition-dependent input $|\phi(c)\rangle$ and is shared across conditions, giving $q_{\boldsymbol{\theta}}(x \mid c) = |\langle x|U_{\boldsymbol{\theta}}|\phi(c)\rangle|^2$. We establish sufficient conditions on the encoder for efficient classical estimation of the conditional training observables and their gradients, and construct encoders satisfying these conditions. Specifically, for the reference amplitudes $\psi_c(z) = \langle z|H^{\otimes n}|\phi(c)\rangle$, we require efficient sampling from $|\psi_c|^2$ and evaluation of amplitude ratios. Building on sparse-observable methods [29, 30] and IQP training methods [28], we derive explicit unbiased estimators of conditional Pauli-Z moments and phase derivatives, with second moments bounded by one and four, respectively. Each specified quantity therefore admits additive error ϵ at fixed confidence using $O(\epsilon^{-2})$ reference samples. We combine these estimators to obtain unbiased estimates of the conditional MMD objective and its gradients from finite samples, enabling classical optimization of a single IQP Born machine across conditions. Figure 1 summarizes the framework.

The numerical experiments cover a translated Gaussian family and temperature-dependent Curie–Weiss distributions at sixteen qubits. A shared model learns translations of the Gaussian output profiles. At temperatures omitted from fitting, the shared Curie–Weiss model has lower mean full-distribution total variation distance (TVD) at $\beta = 1.0$ and at the extrapolation point $\beta = 1.6$ than independent circuits fitted directly at those temperatures. The independent conditional model has lower mean error at $\beta = 0.4$, where the uniform encoder-only baseline has the lowest error in this comparison. Because the models have different fitting roles and aggregate training budgets, these comparisons do not isolate the effect of parameter sharing. Runs from 64 to 2048

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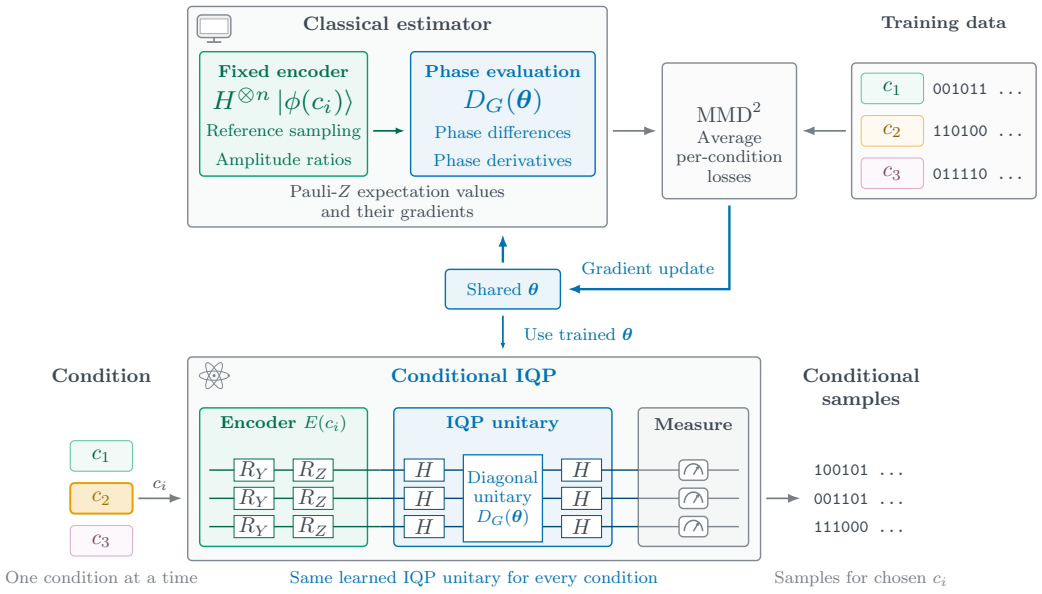


FIG. 1. Proposed framework for classical training and quantum sampling of conditional instantaneous quantum polynomial-time (IQP) Born machines. An encoder maps each condition c_i to an input state $|\phi(c_i)\rangle$. During classical training, samples and amplitude ratios from the reference state $H^{\otimes n}|\phi(c_i)\rangle$, together with phase differences induced by the IQP layer, yield estimates of conditional Pauli-Z moments and their gradients. These estimates enter the squared-MMD objective, averaged across training conditions to optimize one shared parameter vector $\boldsymbol{\theta}$. During quantum sampling, a chosen condition c_i determines the input state. Applying the trained IQP unitary $U_{\boldsymbol{\theta}}$ and measuring in the computational basis produces samples from the learned conditional distribution $q_{\boldsymbol{\theta}}(\cdot \mid c_i)$.

qubits assess computational feasibility by tracking sampled MMD^2 at held-out temperatures throughout optimization.

Furthermore, we show how input-state encoding can enforce exact output symmetries in both conditional and unconditional IQP Born machines. For the Curie–Weiss model, we compare product-state and spin-flip invariant temperature encoders under matched optimization budgets to assess how this constraint affects distributional fit. A fixed-input example illustrates the same symmetry mechanism in an unconditional model.

The remainder of the paper is organized as follows. Section II introduces the conditional IQP model and its classical estimators. Section III develops the conditional MMD training protocol. Section IV describes the numerical experiments and reports their results. Section V shows how input-state preparation can enforce output symmetries. Finally, Sections VI and VII discuss the implications and limitations of our findings and summarize the contributions.

II. CONDITIONAL IQP MODEL

The proposed model separates the dependence on the condition from the learned unitary transformation. An encoder maps each $c \in \mathcal{C}$ to an input state $|\phi(c)\rangle$, while one parameterized IQP unitary $U_{\boldsymbol{\theta}}$ acts at every condition. For a target family $\{p(\cdot \mid c) : c \in \mathcal{C}\}$ over n-bit strings, the model distribution is

$$q_{\boldsymbol{\theta}}(x \mid c) = |\langle x| U_{\boldsymbol{\theta}} |\phi(c)\rangle|^2, \qquad x \in \{0,1\}^n. \tag{1}$$

The encoder determines how the generative model responds to c; the parameterized ansatz determines the transformation shared by the family. Throughout this work, "IQP unitary" refers only to the learned block $U_{\boldsymbol{\theta}}$. The state-preparation circuit need not have IQP form, so the complete conditional circuit need not belong to the strict IQP circuit class. The central question is which condition-dependent input states preserve classical access to the observables and gradients required for training.

The choice of encoder also restricts the conditional functions the model can represent. For rotation-based data encoding, the encoding generators determine accessible Fourier frequencies, while the learned circuit controls their coefficients [31]. Multidimensional Fourier analyses further characterize the dependence on encoding and circuit architecture [32]. These expressivity results motivate treating encoder design separately from the classical estimation criterion.

We parameterize the shared IQP unitary by an undirected graph $G = (V, E)$ with $n = |V|$ vertices. Each vertex $i \in V$ contributes a one-body generator Z_i, and each edge $\{i, j\} \in E$ contributes a two-body generator $Z_i Z_j$. The unitary is

$$U_{\boldsymbol{\theta}} = H^{\otimes n} D_G(\boldsymbol{\theta}) H^{\otimes n}, \tag{2}$$

where

$$D_G(\boldsymbol{\theta}) = \exp\left(i \sum_{i \in V} \theta_i Z_i + i \sum_{\{i,j\} \in E} \theta_{ij} Z_i Z_j \right). \quad (3)$$

Because all Pauli-Z generators commute, the corresponding phase gates constitute a commuting diagonal layer. The evaluated ansätze use at most two-body generators. The phase layer also accommodates k-body Pauli interactions.

Classical training does not require samples from the circuit's output distribution. It instead uses the reference state immediately before the diagonal IQP layer. Define

$$\psi_c(z) = \langle z | H^{\otimes n} | \phi(c) \rangle, \quad (4)$$

so that $|\psi_c(z)|^2$ is the computational-basis distribution of the reference state associated with condition c. We restrict attention to encoders for which, over the relevant conditions c and bit masks a, one can efficiently

1. sample $z \sim |\psi_c(z)|^2$, and
2. evaluate the amplitude ratio

$$R_c(z, a) = \frac{\psi_c(z \oplus a)}{\psi_c(z)} \quad (5)$$

for sampled z.

We refer to these two operations as the sampling-and-ratio interface.

These operations expose the two quantities changed by a Pauli-X mask. Sampling selects a reference-basis string z, while the ratio compares its amplitude with that of the paired string $z \oplus a$. The diagonal IQP layer contributes the phase difference between the same two strings. Their product gives one Monte Carlo contribution to a Pauli-Z moment [29].

The estimator requires the ratio only on the support of $|\psi_c|^2$, since it never samples strings with $\psi_c(z) = 0$. Computationally tractable states in the sense of Van den Nest [29], which admit efficient computational-basis sampling and amplitude evaluation, provide a sufficient realization of this interface under suitable numerical precision assumptions [29].

For a product-state encoder $|\phi(c)\rangle = \bigotimes_{i=1}^n |\phi_i(c)\rangle$, the reference amplitudes factorize as $\psi_c(z) = \prod_i \psi_{c,i}(z_i)$, where $\psi_{c,i}(b) = \langle b | H | \phi_i(c) \rangle$. We can therefore generate reference samples by drawing each bit independently, and the amplitude ratio is

$$R_c(z, a) = \prod_{i : a_i = 1} \frac{\psi_{c,i}(1 - z_i)}{\psi_{c,i}(z_i)}. \quad (6)$$

When the local amplitudes have constant arithmetic evaluation cost, both operations require at most $O(n)$ local operations.

Efficient classical estimation of the observables used for training does not imply efficient classical sampling from $q_{\boldsymbol{\theta}}(\cdot \mid c)$. Furthermore, existing IQP sampling-hardness results apply to specified circuit families and do not establish hardness for arbitrary condition-dependent input states or particular trained parameters. Lastly, practical inference on a quantum computer requires efficient preparation circuits for both $|\phi(c)\rangle$ and the IQP unitary $U_{\boldsymbol{\theta}}$, which the classical sampling-and-ratio interface does not inherently provide.

A. Conditional Pauli estimator and gradients

The conditional MMD objective in Sec. III reduces to Pauli-Z moments. For a mask a, write $Z_a = \bigotimes_{i=1}^n Z_i^{a_i}$ and denote its model expectation by $\mu_q(a \mid c)$. We now show how reference-state access yields an estimator of this quantity and its phase derivatives.

We write the diagonal IQP layer as

$$D_{\boldsymbol{\theta}} |z\rangle = e^{i f_{\boldsymbol{\theta}}(z)} |z\rangle, \quad (7)$$

with

$$f_{\boldsymbol{\theta}}(z) = \sum_{j=1}^{P} \theta_j \chi_{g_j}(z), \qquad \chi_{g_j}(z) = (-1)^{g_j \cdot z}, \quad (8)$$

where P is the number of generators and $g_j \in \{0,1\}^n$ specifies the support of the jth diagonal Pauli generator. The phase change induced by flipping the bits selected by a is

$$\Delta_a f_{\boldsymbol{\theta}}(z) = f_{\boldsymbol{\theta}}(z \oplus a) - f_{\boldsymbol{\theta}}(z). \quad (9)$$

Using the reference amplitudes and amplitude ratio defined in Eqs. (4) and (5), define

$$\begin{aligned} W_{c,a}(z; \boldsymbol{\theta}) &= R_c(z, a) e^{i \Delta_a f_{\boldsymbol{\theta}}(z)}, \\ Y_{c,a}(z; \boldsymbol{\theta}) &= \mathrm{Re}\, W_{c,a}(z; \boldsymbol{\theta}). \end{aligned} \quad (10)$$

The derivative of the phase difference is

$$d_{j,a}(z) = \chi_{g_j}(z \oplus a) - \chi_{g_j}(z) \in \{-2, 0, 2\}. \quad (11)$$

The following proposition specializes sparse-observable amplitude-ratio estimation [30] to conditional IQP moments and their phase derivatives. For computationally tractable reference states, efficient estimation also follows from Van den Nest's sparse-operator theorem [29].

Proposition 1 (Classical estimation from reference-state access)**.** *Fix a condition c, a mask a, and a phase function with polynomially many diagonal Pauli terms as in Eq. (8). Suppose that the normalized reference state is independent of $\boldsymbol{\theta}$, and that independent sampling from $|\psi_c|^2$ and evaluation of $R_c(z, a)$ on its sampled support are efficient. For $z \sim |\psi_c|^2$, with exact sampling and arithmetic, the random variables*

$$\begin{aligned} Y &= \mathrm{Re}\, W_{c,a}(z; \boldsymbol{\theta}), \\ Y'_j &= \mathrm{Re}[i\, d_{j,a}(z) W_{c,a}(z; \boldsymbol{\theta})] \end{aligned} \quad (12)$$

satisfy

$$\begin{aligned} \mathbb{E}[Y] &= \mu_q(a \mid c), & \mathbb{E}[Y^2] &\le 1, \\ \mathbb{E}[Y'_j] &= \partial_{\theta_j}\mu_q(a \mid c), & \mathbb{E}[(Y'_j)^2] &\le 4. \end{aligned} \tag{13}$$

The means of S independent samples are therefore unbiased, with mean-square errors at most $1/S$ and $4/S$, respectively. Each specified moment or phase derivative can be estimated to additive error ϵ with fixed confidence using $S = O(\epsilon^{-2})$ samples. Full support is not required.

Proof. Let $S_c = \{z : \psi_c(z) \neq 0\}$. Conjugating the measured observable through the final Hadamard layer gives $X_a = \bigotimes_{i=1}^n X_i^{a_i}$, so

$$\begin{aligned} \mu_q(a \mid c) &= \langle \psi_c | \, D_{\boldsymbol{\theta}}^\dagger X_a D_{\boldsymbol{\theta}} \, | \psi_c \rangle \\ &= \mathrm{Re} \sum_{z \in S_c} \psi_c(z)^* \psi_c(z \oplus a) e^{i \Delta_a f_{\boldsymbol{\theta}}(z)} \\ &= \mathbb{E}_{z \sim |\psi_c|^2}[Y]. \end{aligned} \tag{14}$$

Because the reference state is independent of $\boldsymbol{\theta}$, differentiating the finite sum gives

$$\begin{aligned} \partial_{\theta_j}\mu_q(a \mid c) &= \mathbb{E}_{z \sim |\psi_c|^2}[\mathrm{Re}\,(i\, d_{j,a}(z) W_{c,a}(z;\boldsymbol{\theta}))] \\ &= \mathbb{E}[Y'_j]. \end{aligned} \tag{15}$$

Although individual amplitude ratios can be large, normalization bounds their second moment:

$$\mathbb{E}[|W_{c,a}|^2] = \sum_{z \in S_c} |\psi_c(z \oplus a)|^2 \le 1. \tag{16}$$

Thus $\mathbb{E}[Y^2] \le 1$. Since $|d_{j,a}(z)| \le 2$, we also obtain $\mathbb{E}[(Y'_j)^2] \le 4$. The means of S independent samples consequently have mean-square errors at most $1/S$ and $4/S$. Chebyshev's inequality gives the stated fixed-confidence sample complexity. Full support is unnecessary because the estimator evaluates ratios only on the sampled support. □

In practice, we estimate the moment using independent reference samples $z_s \sim |\psi_c|^2$,

$$\widehat{\mu}_q(a \mid c) = \frac{1}{S} \sum_{s=1}^{S} Y_{c,a}(z_s;\boldsymbol{\theta}). \tag{17}$$

Averaging the corresponding Y'_j values over the same samples estimates the phase derivative. For the standard IQP input, the reference amplitudes are uniform and $R_c(z,a) = 1$, so Eq. (14) reduces to the usual cosine estimator [29, 33].

For a circuit with r ancillary qubits discarded at measurement, the same construction uses the joint reference state and replaces each measured mask a by $(a, 0^r)$.

III. CONDITIONAL IQP TRAINING

This section formalizes the training of conditional IQP Born machines. Throughout this work, classically trainable refers to efficient classical estimation of the training objective and its phase gradients to additive accuracy.

A. Condition-balanced MMD objective

We optimize the shared parameter vector $\boldsymbol{\theta}$ by minimizing the squared maximum mean discrepancy (MMD) across the training conditions. For distributions p and q on $\{0,1\}^n$ and a positive-definite kernel k, the population squared MMD is [26]

$$\begin{aligned} \mathrm{MMD}^2(p,q) = \mathbb{E}_{x,x' \sim p}[k(x,x')] - 2\mathbb{E}_{x \sim p,\, y \sim q}[k(x,y)] \\ + \mathbb{E}_{y,y' \sim q}[k(y,y')], \end{aligned} \tag{18}$$

where each expectation uses independent paired draws.

Throughout this work, we use the Gaussian (Hamming) kernel

$$k_\sigma(x,y) = \exp\left[-\frac{d_H(x,y)}{2\sigma^2}\right], \tag{19}$$

where d_H is the Hamming distance and $\sigma > 0$ is the kernel bandwidth.

Let $\mathcal{C}_{\mathrm{train}}$ denote the set of training conditions and τ the set of kernel bandwidths. We optimize the model with the condition-balanced objective

$$\mathcal{L}(\boldsymbol{\theta}) = \frac{1}{|\mathcal{C}_{\mathrm{train}}|} \sum_{c \in \mathcal{C}_{\mathrm{train}}} \frac{1}{|\tau|} \sum_{\sigma \in \tau} \mathrm{MMD}^2_\sigma(p(\cdot \mid c), q_{\boldsymbol{\theta}}(\cdot \mid c)). \tag{20}$$

The objective weights all training conditions and bandwidths equally. At each optimizer update, we independently sample one condition and one bandwidth uniformly, giving a stochastic estimate of this average.

B. Character expansion

For binary strings, the Gaussian kernel admits an exact expansion in binary characters. Let $a \in \{0,1\}^n$ be a random visible-bit mask with independent coordinates

$$a_i \sim \mathrm{Bernoulli}(p_\sigma), \qquad p_\sigma = \frac{1 - e^{-1/(2\sigma^2)}}{2}, \tag{21}$$

and define the binary character

$$\chi_a(x) = (-1)^{a \cdot x}. \tag{22}$$

The kernel can then be written as

$$k_\sigma(x,y) = \mathbb{E}_a[\chi_a(x)\chi_a(y)]. \tag{23}$$

For a condition c, define the target and model character moments

$$\begin{aligned} \mu_p(a \mid c) &= \mathbb{E}_{x \sim p(\cdot|c)}[\chi_a(x)], \\ \mu_q(a \mid c) &= \mathbb{E}_{x \sim q_{\boldsymbol{\theta}}(\cdot|c)}[\chi_a(x)]. \end{aligned} \tag{24}$$

Substituting Eq. (23) into Eq. (18) gives

$$\mathrm{MMD}^2_\sigma(p(\cdot \mid c), q_{\boldsymbol{\theta}}(\cdot \mid c)) = \mathbb{E}_a\left[(\mu_p(a \mid c) - \mu_q(a \mid c))^2\right]. \tag{25}$$

For a Born machine, $\chi_a(x)$ is the eigenvalue of the visible Pauli operator

$$Z_a = \bigotimes_{i=1}^{n} Z_i^{a_i}, \tag{26}$$

and therefore

$$\mu_q(a \mid c) = \langle Z_a \rangle_{q_{\boldsymbol{\theta}}(\cdot|c)}. \tag{27}$$

We can thus express the conditional MMD entirely in terms of visible Pauli-Z moments. This representation allows classical training to use the estimator of IQP expectation values detailed in Sec. II A.

The bandwidth σ determines which correlation orders are weighted most strongly. Since the mask coordinates are sampled independently, its Hamming weight $K = |a|$ follows

$$\begin{aligned} K &\sim \text{Binomial}(n, p_\sigma), \\ \mathbb{E}[K] &= np_\sigma, \\ \text{Var}(K) &= np_\sigma(1-p_\sigma). \end{aligned} \tag{28}$$

Thus, each bandwidth induces a distribution over Pauli-string weights rather than selecting a single correlation order. Large bandwidths favor low-weight observables, whereas decreasing σ shifts weight toward higher-order correlations, with $\mathbb{E}[K] \to n/2$ as $\sigma \to 0$.

The bandwidth controls which correlations the objective emphasizes and can affect the optimization landscape. Efficient classical estimation does not guarantee that optimization finds an accurate model. Prior work shows that global cost terms can induce barren plateaus [34, 35], including in shallow circuits [36]. Restricting the objective to low-body correlations may mitigate these difficulties, but reduces its sensitivity to distributions that differ in higher-order correlations [37].

C. Unbiased finite-sample MMD and gradients

For a fixed condition c and visible mask a, let

$$T_b = \chi_a(x_b), \qquad x_b \sim p(\cdot \mid c), \tag{29}$$

for $B \geq 2$ independent target samples, and let

$$Y_s = Y_{c,a}(z_s; \boldsymbol{\theta}), \qquad z_s \sim |\psi_c|^2, \tag{30}$$

for $S \geq 2$ independent reference samples drawn independently of the target samples, with $Y_{c,a}$ defined in Eq. (10). Define the corresponding sample means

$$\overline{T} = \frac{1}{B}\sum_{b=1}^{B} T_b, \qquad \overline{Y} = \frac{1}{S}\sum_{s=1}^{S} Y_s. \tag{31}$$

A direct estimate of the squared moment difference, $(\overline{Y} - \overline{T})^2$, is biased because squaring a finite-sample mean introduces its sampling variance. We remove this bias by forming products only between independent sample pairs,

$$\begin{aligned} \widehat{\Delta_c^2}(a) = &\frac{1}{S(S-1)}\sum_{s\neq t} Y_s Y_t - 2\overline{Y}\,\overline{T} \\ &+ \frac{1}{B(B-1)}\sum_{b\neq b'} T_b T_{b'}. \end{aligned} \tag{32}$$

Independence of the paired samples gives

$$\mathbb{E}\left[\widehat{\Delta_c^2}(a)\right] = \left(\mu_q(a \mid c) - \mu_p(a \mid c)\right)^2. \tag{33}$$

Thus, Eq. (32) provides an unbiased finite-sample estimator of the squared character-moment difference appearing in Eq. (25). An unbiased finite-sample estimate of MMD^2 may be negative.

The pair sums have the equivalent forms

$$\sum_{s\neq t} Y_s Y_t = \left(\sum_s Y_s\right)^2 - \sum_s Y_s^2, \tag{34}$$

and, since $T_b^2 = 1$,

$$\sum_{b\neq b'} T_b T_{b'} = \left(\sum_b T_b\right)^2 - B. \tag{35}$$

We can therefore evaluate the estimator in linear rather than quadratic time in the numbers of target and reference samples.

The same construction gives an unbiased estimator of the phase gradients. Let

$$Y_s' = \partial_{\theta_j} Y_{c,a}(z_s; \boldsymbol{\theta}), \tag{36}$$

with Y_s' evaluated using Eq. (15). Differentiating Eq. (32) gives

$$\partial_{\theta_j}\widehat{\Delta_c^2}(a) = \frac{2}{S(S-1)}\sum_{s\neq t} Y_s Y_t' - 2\overline{T}\,\overline{Y'}, \tag{37}$$

where

$$\overline{Y'} = \frac{1}{S}\sum_s Y_s'. \tag{38}$$

Taking the expectation yields

$$\mathbb{E}\left[\partial_{\theta_j}\widehat{\Delta_c^2}(a)\right] = 2\left(\mu_q(a \mid c) - \mu_p(a \mid c)\right)\partial_{\theta_j}\mu_q(a \mid c), \tag{39}$$

which is the gradient of the corresponding squared moment difference.

At a fixed parameter vector, these population unbiasedness statements average over independent target and reference samples. During training, we sample target minibatches with replacement from fixed training pools; the stochastic gradient is therefore unbiased for the corresponding empirical objective.

Excluding the diagonal terms $s = t$ removes products of Y_s and Y'_s computed from the same reference draw, which can be correlated. Since the encoder is fixed, the reference sampling law does not depend on $\boldsymbol{\theta}$; consequently, holding the sampled reference strings fixed during automatic differentiation evaluates the required phase gradients without differentiating through the sampler.

D. Stochastic mask estimator

Equation (25) expresses the MMD as an expectation over binary masks a. Since the number of possible masks grows exponentially with n, we estimate this expectation by drawing a finite sample of masks from the distribution in Eq. (21).

The identity mask $a = 0$ satisfies

$$\mu_p(0 \mid c) = \mu_q(0 \mid c) = 1 \tag{40}$$

for every condition and therefore contributes identically zero to the MMD. We consequently sample only nonidentity masks from the conditional distribution

$$P_\sigma(a \mid a \neq 0). \tag{41}$$

The total probability assigned to nonidentity masks under the original kernel distribution is

$$Q_\sigma^+ = 1 - (1 - p_\sigma)^n. \tag{42}$$

Multiplying the conditional expectation by Q_σ^+ therefore preserves the original Gaussian kernel objective.

For M sampled masks $a_1, \ldots, a_M$, the finite-mask estimator at condition c is

$$\widehat{\mathrm{MMD}}^2_\sigma(c) = Q_\sigma^+ \frac{1}{M} \sum_{m=1}^{M} \widehat{\Delta^2_c}(a_m), \tag{43}$$

where $\widehat{\Delta^2_c}$ is the unbiased finite-sample estimator in Eq. (32). Averaging also over the mask draws recovers the full kernel MMD.

We need not evaluate repeated masks more than once. If $\mathcal{U}$ denotes the set of distinct sampled masks and m_u the multiplicity of mask u, Eq. (43) can be written equivalently as

$$\widehat{\mathrm{MMD}}^2_\sigma(c) = Q_\sigma^+ \sum_{u \in \mathcal{U}} \frac{m_u}{M} \widehat{\Delta^2_c}(u). \tag{44}$$

Combining duplicate masks, therefore, reduces the number of Pauli moments we must evaluate without changing either the sampled objective or its gradient.

IV. NUMERICAL EXPERIMENTS

We evaluate the proposed architecture on a translated Gaussian family and the Curie–Weiss model, with a supplementary scaling demonstration up to 2048 qubits. For the two sixteen-qubit benchmarks, the primary accuracy metric is the exact full-distribution total variation distance,

$$\mathrm{TVD}(p, q) = \frac{1}{2} \sum_{x \in \{0,1\}^n} |p(x) - q(x)|. \tag{45}$$

We also report the exact Hamming-profile TVD as a diagnostic,

$$\mathrm{TVD}_K(p, q) = \frac{1}{2} \sum_{k=0}^{n} |\rho_p(k) - \rho_q(k)|, \tag{46}$$

where $\rho(k)$ is the probability of observing a bitstring of weight k. Aggregation gives $\mathrm{TVD}_K(p, q) \leq \mathrm{TVD}(p, q)$. The targets are uniform within each Hamming-weight shell, but the learned models need not be, so accurate profiles can hide distributional errors. Both metrics use the same exact circuit probabilities.

Training minimizes the squared-MMD objective. Gaussian checkpoints use development-profile TVD, whereas all sixteen-qubit Curie–Weiss models use full-distribution TVD for hyperparameter and checkpoint selection. Appendix A specifies the evaluation protocols.

A. Discrete Gaussian translation

Our first experiment illustrates the conditioning mechanism with a moving Gaussian profile. For $n = 16$ bits, let $K = |x|$ be the Hamming weight. The target distribution translates the nominal center of a discrete Gaussian based on a condition $c \in [0.0, 1.0]$, with all bitstrings of equal weight having equal target probability.

We use a fixed product-state encoder that applies the same two rotations to every qubit:

$$|\phi(c)\rangle = \bigotimes_{i=1}^{16} R_Z\left(c - \frac{1}{2}\right) R_Y(1.5)\, |0\rangle_i \,. \tag{47}$$

Measurement immediately after this encoder yields a fixed distribution; changing c modifies the input phases but leaves the outcome probabilities unchanged.

The shared unitary uses a 4×4 lattice with 16 one-body gates and 24 horizontal and vertical nearest-neighbor gates, giving 40 trainable parameters.

We train at $c \in \{0.0, 0.25, 0.50, 0.75, 1.0\}$ and evaluate across 80 held-out conditions. Across five runs, the median full TVD averaged over evaluation conditions is 0.0995, compared with 0.2922 for the encoder alone (Table I). The corresponding median profile TVD is 0.0464; the larger full TVD reveals errors hidden by Hamming-weight aggregation. Figure 2 shows the translated profiles and conditional response. Appendix A 2 gives the training and evaluation protocol.

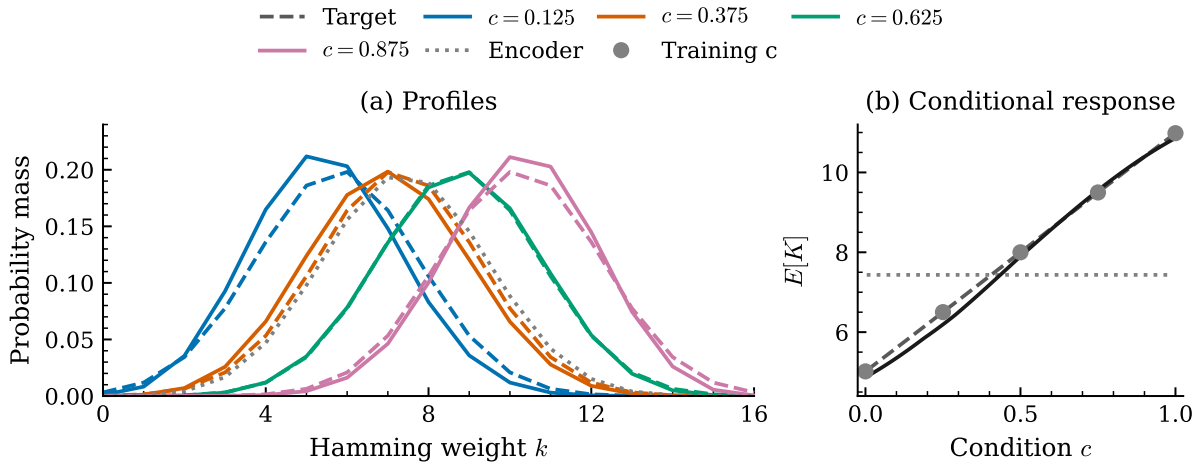


FIG. 2. Mean Hamming-weight translation via conditioning at sixteen qubits. (a) Solid colored curves show mean exact model profiles across five runs at the midpoint development conditions $c \in \{0.125, 0.375, 0.625, 0.875\}$, which are excluded from fitting but used for checkpoint selection. (b) The solid black conditional-response line shows the model's mean output Hamming weight at each of the 80 held-out conditions, summarized by the pointwise median across runs; shading shows the interquartile range. Dashed curves show exact targets, dotted gray curves show the encoder alone, and gray circles mark training conditions on the target response. Table I reports exact full-distribution and profile errors.

TABLE I. Exact Gaussian full-distribution and secondary profile TVD. Each value is the median across 5 runs of the mean error over 80 held-out conditions. Selection uses development-profile TVD; bold entries mark the lowest value in each column.

Model	Full TVD ↓	Profile TVD ↓
Conditional IQP	**0.0995**	**0.0464**
Encoder only	0.2922	0.2922

B. Curie-Weiss spin model

The Curie–Weiss model [38] tests the architecture's ability to capture changes in distributional shape. The spins $s_i \in \{-1, +1\}$ interact with equal strength, so the target assigns equal probability to bitstrings with the same Hamming weight. As the inverse temperature β increases, the finite-size Hamming-weight profile changes from a central peak to two symmetric modes.

We encode the inverse temperature using the fixed product phase encoder

$$|\phi(\beta)\rangle = \bigotimes_{i=1}^{16} R_Z(0.75\beta) R_Y\left(\frac{\pi}{2}\right) |0\rangle_i \,. \tag{48}$$

The encoder-only output is uniform for every β. The parameterized circuit uses all one- and two-body generators, giving 136 parameters. We train at $\beta \in \{0.6, 0.8, 1.2, 1.4\}$ and evaluate at $\beta \in \{0.4, 1.0, 1.6\}$.

Two independently tuned baseline families provide comparisons with the shared conditional IQP model. Each family fits one circuit at each reporting temperature. Independent conditional IQPs retain the corresponding temperature encoder, whereas independent unconditional IQPs use $|0\rangle^{\otimes 16}$.

Exact full-distribution TVD determines checkpoint selection and hyperparameter optimization (HPO). The shared model uses development temperatures, while each independent family selects its configuration jointly across its fitted temperatures. Appendix A 3 a gives the protocol.

Figure 3 compares the magnetization profiles, while Table II reports errors over the full distribution. The shared conditional model has the lowest mean error among trained families at $\beta = 1.0$ and 1.6, with mean errors 0.2175 and 0.1709, respectively. The independent conditional model has lower mean error at $\beta = 0.4$, 0.1240 compared with 0.2240 for the shared model. At this temperature, the uniform encoder-only law has error 0.1180, lower than every trained family in this comparison. Relative to that encoder, the shared model improves mean full TVD at $\beta = 1.0$ and 1.6 but worsens it at $\beta = 0.4$. These findings describe this target family and the specified full-TVD selection protocol; they do not establish a general advantage of parameter sharing.

Each independent family uses three circuits, with three times the total number of parameters and optimizer updates of the shared model. The shared model uses more total training-pool samples. These unequal aggregate resources and the different evaluation roles limit conclusions about parameter sharing alone. Appendix A 3 reports the detailed budgets.

Table VII in Appendix A 3 reports the secondary profile diagnostic, which gives a different ranking. Both independent models have lower mean profile TVD at $\beta = 1.0$, despite higher full TVD than the shared model. The

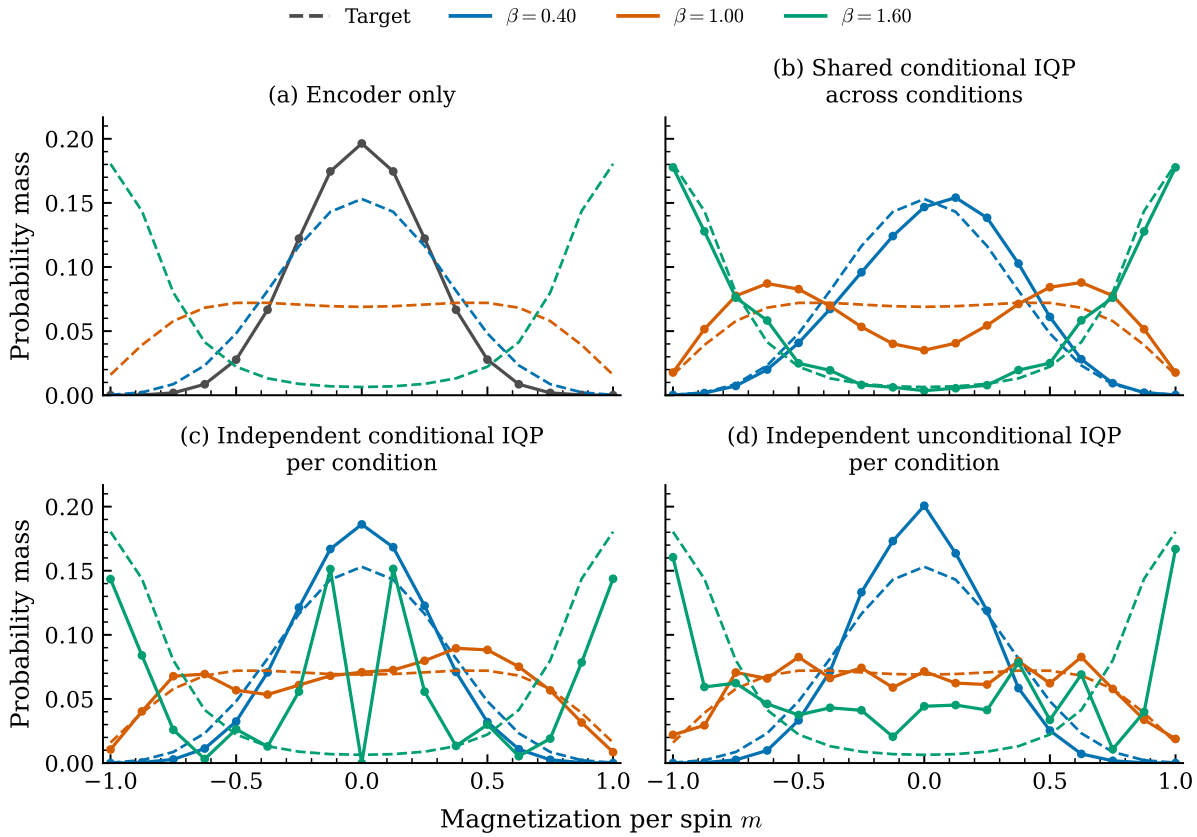


FIG. 3. Curie–Weiss magnetization profiles at sixteen qubits, with $m = (2k - 16)/16$. Panels show (a) encoder only, (b) shared conditional IQP, (c) independent conditional IQPs, and (d) independent unconditional IQPs. Colors identify $\beta = 0.4, 1.0, 1.6$; dashed curves show exact targets and solid curves show mean model profiles over five runs from the confirmation cohort used for model selection. The encoder is deterministic. Shared reporting temperatures are excluded from fitting and current selection, while independent circuits fit at each displayed temperature. Table II reports full-distribution accuracy.

TABLE II. Exact full-distribution TVD for sixteen-qubit Curie–Weiss models. Trained entries give means ± sample standard deviations across 5 runs from the confirmation cohort used for model selection, reused for reporting; the encoder is deterministic. Shared reporting temperatures are excluded from fitting and current selection, while independent circuits fit at each reporting temperature. Bold entries mark the lowest mean error, including the encoder.

Model	Evaluation	Full-distribution TVD ↓		
		$\beta = 0.4$	$\beta = 1.0$	$\beta = 1.6$
Encoder only	Baseline	**0.1180**	0.4406	0.8699
Shared conditional IQP	Held-out	0.2240 ± 0.0115	$\mathbf{0.2175 \pm 0.0136}$	$\mathbf{0.1709 \pm 0.0136}$
Independent conditional IQPs	Fitted	0.1240 ± 0.0083	0.2309 ± 0.0319	0.3948 ± 0.0003
Independent unconditional IQPs	Fitted	0.3842 ± 0.0017	0.3322 ± 0.0105	0.3299 ± 0.0119

shared model also improves the encoder's profile TVD at $\beta = 0.4$, while worsening full TVD. These differences show why profile agreement alone does not establish accurate generation of bitstrings.

1. *Spin-flip invariant temperature encoding*

The zero-field Curie–Weiss target is invariant under reversal of all spins, $p(x \mid \beta) = p(\bar{x} \mid \beta)$. We impose this constraint through the input state while retaining the temperature map and all 136 independent circuit phases. Writing $t = 0.75\beta$ and $F = X^{\otimes n}$, we project the product

encoder onto the +1 eigenspace of F:

$$|\phi_+(\beta)\rangle = \frac{(I+F)\,|\phi(\beta)\rangle}{\sqrt{2[1+\cos^n(t)]}}. \tag{49}$$

The normalization follows from $\langle\phi(\beta)|F|\phi(\beta)\rangle = \cos^n(t)$. The learned IQP unitary commutes with F, so $FU_{\boldsymbol{\theta}}\,|\phi_+(\beta)\rangle = U_{\boldsymbol{\theta}}\,|\phi_+(\beta)\rangle$ and the output satisfies $q_{\boldsymbol{\theta},+}(x \mid \beta) = q_{\boldsymbol{\theta},+}(\bar{x} \mid \beta)$ for every parameter vector. This conditional input applies the commuting-symmetry construction of Recio-Armengol *et al.* [28, Sec. 6], which we discuss in Sec. V. Independent learned phases can still break permutation symmetry, so spin-flip invariance does not imply uniform probabilities within Hamming shells.

The projected encoder preserves efficient classical access to the reference state. For $k = |z|$,

$$\psi_{+,\beta}(z) = \frac{\mathbf{1}_{k\ \mathrm{even}}[\cos(t/2)]^{n-k}[-i\sin(t/2)]^k}{\sqrt{[1+\cos^n(t)]/2}}. \tag{50}$$

We sample k from a binomial law with success probability $\sin^2(t/2)$ conditioned on even parity, then choose a uniformly random weight-k bitstring. For a sampled z, the amplitude ratio is

$$R_{+,\beta}(z,a) = \begin{cases} [-i\tan(t/2)]^{|a|-2z\cdot a}, & |a| \text{ even}, \\ 0, & |a| \text{ odd}. \end{cases} \tag{51}$$

These operations require polynomial resources and enter the same conditional estimator and training objective. A deterministic preparation circuit uses a running parity wire within the existing register and $O(n)$ gates, without an ancillary qubit or postselection.

The encoder comparison uses separate HPO studies with matched budgets and full-distribution TVD for selection. We evaluate the models using the confirmation cohort that selected the winning configuration. Both models use the same fitting and development temperatures and training budgets. Appendix A gives the specific configurations and comparison protocol. Table III reports exact full-distribution TVD. We quantify deviation from global spin-flip symmetry by

$$D_{\mathrm{flip}}(q \mid \beta) = \frac{1}{2}\sum_x |q(x \mid \beta) - q(\bar{x} \mid \beta)|\,. \tag{52}$$

Here $\bar{x}$ complements every bit of x, and $D_{\mathrm{flip}} = 0$ if and only if the output distribution is spin-flip invariant. For the trained product model, mean D_{flip} is 0.4354, 0.1092, and 0.0087 at $\beta = 0.4, 1.0, 1.6$, respectively. The invariant input enforces $D_{\mathrm{flip}} = 0$ at every temperature, independently of optimization; numerical evaluations confirm this equality to machine precision. Distributional fit remains an empirical question.

The invariant input lowers mean full-distribution TVD at $\beta = 0.4$ and 1.0, while the product input has lower mean error at $\beta = 1.6$ (Table III). At the two lower temperatures, the improvement holds across all paired runs; at the highest temperature, its direction varies across runs. This exact symmetry does not enforce uniform probabilities within Hamming shells, and appreciable full-distribution errors remain.

Appendix A 3 b reports encoder-only errors and paired full-TVD differences. Input projection enforces symmetry exactly, but its effect on distributional accuracy depends on temperature.

C. Scaling

The scaling study assesses the computational feasibility of classical training at $n \in \{64, 128, 256, 512, 1024, 2048\}$ visible qubits, without ancillas. Each qubit receives the fixed product phase encoder $R_Z(0.75\beta)R_Y(\pi/2)$. The parameterized unitary contains independent one-body phases and pair phases between horizontal and vertical nearest neighbors on an open rectangular lattice. For an $L_x \times L_y$ lattice, the ansatz has $3n - L_x - L_y$ parameters, so the parameter count grows linearly with the number of qubits. Appendix A lists the lattice dimensions and optimization hyperparameters.

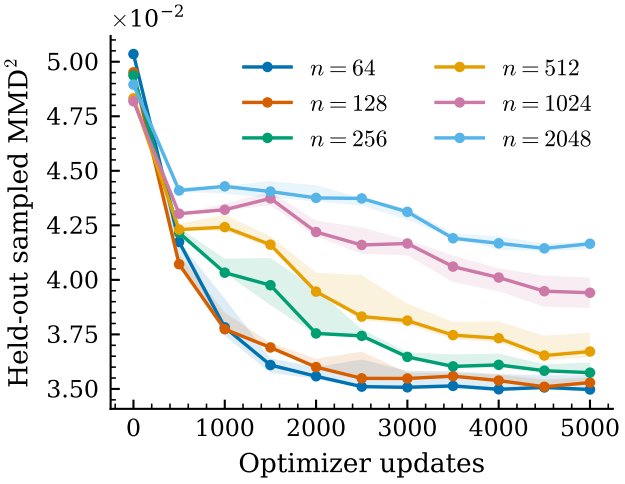


FIG. 4. Classical training of conditional IQP Born machines from 64 to 2048 qubits. Lines and shaded bands show the median and interquartile range of held-out sampled MMD^2 across three independent training runs. Evaluation conditions are $\beta \in \{0.4, 0.95, 1.0, 1.05, 1.6\}$. We apply the same expected mask weights $w \in \{2, 4\}$ at every system size, so the kernel bandwidths vary with system size.

Training uses the same temperatures as the sixteen-qubit shared model. Figure 4 averages sampled MMD^2 across held-out temperatures throughout optimization. Appendix A 4 specifies the training and reporting protocol.

Figure 4 shows a decrease in held-out sampled MMD^2 at every system size, followed by a persistent residual. At 64 qubits, the median across runs decreases from 0.0504 at initialization to 0.0350 after 5000 updates; at 2048

TABLE III. Exact full-distribution TVD for the Curie–Weiss encoder comparison. Values are means ± sample standard deviations over 5 paired runs from the confirmation cohort used for model selection, reused for reporting. Bold entries mark the lowest mean error at each temperature.

Model	$\beta = 0.4$	$\beta = 1.0$	$\beta = 1.6$
Product-state input	0.2240 ± 0.0115	0.2175 ± 0.0136	$\mathbf{0.1709 \pm 0.0136}$
Spin-flip invariant input	$\mathbf{0.0902 \pm 0.0184}$	$\mathbf{0.1889 \pm 0.0134}$	0.1805 ± 0.0276

qubits, it decreases from 0.0490 to 0.0417. These results demonstrate the feasibility of classically optimizing conditional IQP Born machines up to 2048 qubits. The scaling study uses a sparse ansatz and optimizer settings that differ from the sixteen-qubit comparison; transfer of that tuned configuration remains untested. A decrease in sampled MMD^2 does not establish accurate recovery of the full target distribution. Because we apply the same expected mask weights at every size, the corresponding kernel bandwidths vary with system size. Consequently, readers should not interpret absolute MMD^2 values as direct comparisons of distributional fidelity across sizes.

V. INPUT-STATE INDUCED SYMMETRIES IN IQP BORN MACHINES

Encoder design determines both the dependence on a condition and the constraints satisfied by the output law. The projected Curie–Weiss input in Sec. IV B 1 combines these roles. More generally, if a bit-flip operator X_s commutes with the learned unitary and each encoded input is an eigenstate of X_s, the conditional output is invariant under that flip at every condition and parameter vector. We illustrate the same construction with a fixed unconditional input while preserving classical trainability. Such constraints provide inductive biases in variational quantum models [39].

Choose any nonzero bit mask $s \in \{0,1\}^n$ and prepare

$$|\phi_s\rangle = \frac{|0^n\rangle + |s\rangle}{\sqrt{2}}. \tag{53}$$

This state satisfies $X_s |\phi_s\rangle = |\phi_s\rangle$, where $X_s = \bigotimes_{i=1}^n X_i^{s_i}$ flips the qubits selected by s. Since the IQP unitary commutes with X_s, its output remains invariant under the same flip, even when all learned phases vary independently. Consequently,

$$q_{\boldsymbol{\theta}}(x) = q_{\boldsymbol{\theta}}(x \oplus s) \qquad \text{for every } x \text{ and } \boldsymbol{\theta}. \tag{54}$$

The reference amplitudes are uniform on the strings satisfying $s \cdot z = 0$ and vanish elsewhere, evaluating the dot product modulo two. Sampling uniformly subject to this single parity constraint is efficient. For a sampled string z, the amplitude ratio is

$$R_s(z,a) = \begin{cases} 1, & s \cdot a = 0, \\ 0, & s \cdot a = 1. \end{cases} \tag{55}$$

The encoded input therefore satisfies the sampling-and-ratio interface, preserving classical access to the moments and gradients required for training.

For example, $s = 110$ on three qubits gives $|\phi_s\rangle = (|000\rangle + |110\rangle)/\sqrt{2}$ and enforces $q_{\boldsymbol{\theta}}(x) = q_{\boldsymbol{\theta}}(x \oplus 110)$. The ordinary $|000\rangle$ input does not enforce this relation: with only $\theta_1 = \pi/6$ nonzero, it gives $q(000) = 3/4$ and $q(110) = 0$. Figure 5 compares the two inputs under the same unitary. This selected-subset construction illustrates the commuting-symmetry recipe of Recio-Armengol *et al.*, whose demonstrated example enforces global bit-flip symmetry [28].

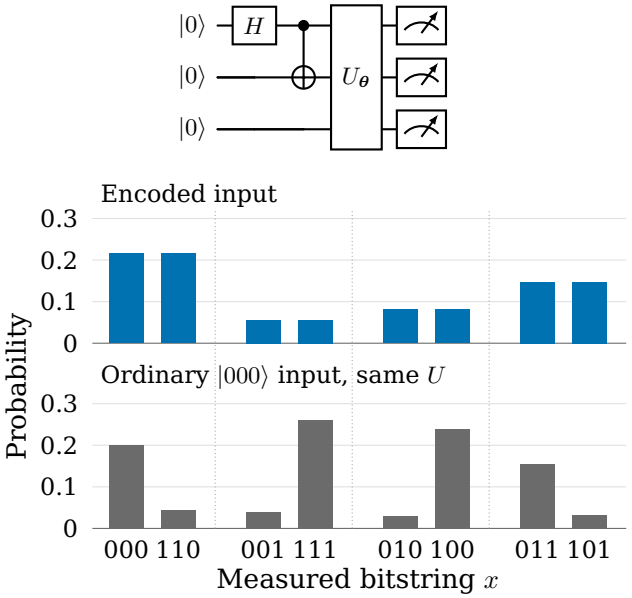


FIG. 5. The encoded input makes paired outcomes equally likely. The circuit prepares $(|000\rangle + |110\rangle)/\sqrt{2}$ before applying $U_{\boldsymbol{\theta}}$. The plots show exact probabilities at one parameter setting for the encoded input (blue) and the ordinary $|000\rangle$ input (gray), using the same unitary in both cases. Adjacent outcomes differ by 110. Their equality for the encoded input holds for every phase vector.

VI. DISCUSSION

Our numerical experiments demonstrate conditional generation with measurable residual full-distribution errors. In the Gaussian experiment, the shared model re-

duces aggregate full TVD relative to the encoder. In the Curie–Weiss baseline comparison, the shared model has lower mean full TVD than both independently fitted families at $\beta = 1.0$ and 1.6. At $\beta = 0.4$, the uniform encoder-only law has lower error than the shared product model and both independent families. Different evaluation roles and unequal aggregate training budgets prevent attributing these outcomes to parameter sharing alone.

Full-distribution TVD exposes errors that Hamming-weight aggregation can hide and changes some model rankings. Efficient estimation of the training moments therefore does not guarantee accurate recovery of the full distribution. Herrero-Gonzalez *et al.* analyze discrepancies between classical correlator-based surrogates and deployed Born distributions [40]. Under the stated sampling and arithmetic assumptions, our estimators target the moments of the specified circuit without truncating Pauli propagation. Nevertheless, finite sampling budgets and the correlations emphasized by the kernel can leave distributional errors unresolved.

Model selection also limits the interpretation of run variability. Gaussian checkpoints use profile TVD for selection, whereas all sixteen-qubit Curie–Weiss models use full-distribution TVD. For Curie–Weiss reporting, we reuse the confirmation cohorts that selected the winning configurations. The reported variability therefore does not constitute an independent assessment of hyperparameter selection. The current protocol excludes shared reporting temperatures from fitting and selection, although earlier experiments examined them.

The scaling experiment demonstrates the computational feasibility of classical training up to 2048 qubits for the tested sparse ansatz. Residual sampled MMD^2 errors remain, and the size-dependent kernels preclude direct comparisons of distributional fidelity across sizes. Diagnostics beyond sampled MMD are needed to assess conditional accuracy at larger sizes.

Encoder design controls the conditional functions available to the model and can impose exact constraints on its output. Both the product phase encoder and its spin-flip invariant projection admit efficient state preparation, reference sampling and amplitude-ratio evaluation. The Curie–Weiss construction in Sec. IV B 1 and the unconditional illustration in Sec. V restrict the output family through the input, independently of the learned phases. In the matched Curie–Weiss comparison, projection lowers mean full TVD relative to the trained product-input model at $\beta = 0.4$ and 1.0, but raises it at $\beta = 1.6$. These empirical results do not follow from the symmetry guarantee, and spin-flip invariance does not resolve all within-shell errors. Classical postprocessing that averages the output distribution with its bitwise complement also enforces spin-flip symmetry and may improve accuracy, since it cannot increase full TVD to an invariant target.

More structured encoders could offer richer conditional dependence if they retain the same reference interface. Spectral Born machines extend IQP generative models through group Fourier analysis while preserving efficient classical training [41]. Investigating conditional reference-state access in that setting could broaden the framework to other forms of discrete data.

Generator selection and restricted initializations [42, 43], together with kernel-adaptive IQP training [44], offer directions for further study. Their benefit for conditional generation remains to be evaluated.

Quantum deployment requires assessing how noise and hardware constraints affect conditional accuracy. Ryu *et al.* propose logical-noise injection to decompose large IQP circuits into smaller clusters that fit qubit-limited processors [45]. This approximate strategy introduces a further potential discrepancy from the trained distribution. Its suitability for conditional models requires evaluating such errors across conditions.

Whether useful trained conditional models retain classical sampling hardness remains open. Herbst *et al.* identify average-case trainability limitations for anticoncentrating output families [46]. Bläser *et al.* propose a cross-entropy verification protocol for cubic IQP circuits with planted independent spaces [47]. Under two complexity conjectures, secret knowledge of the planted structure creates a polynomial gap between verification and classical spoofing costs, although both remain exponential. These guarantees concern a prescribed circuit family and do not establish hardness for our trained conditional models. Practical quantum advantage would require evidence of useful conditional generation together with an appropriate classical comparison and a justified sampling-hardness claim.

VII. CONCLUSIONS

We introduced a conditional formulation for IQP Born machines that combines a condition-dependent encoder with a learned unitary shared across conditions. Efficient reference-state sampling and amplitude-ratio evaluation enable unbiased classical estimation of conditional Pauli-Z moments and phase gradients, supporting optimization without quantum circuit evaluations. The same interface also accommodates input states that impose exact output symmetries.

The Gaussian experiment demonstrates improved full-distribution accuracy relative to the encoder. The Curie–Weiss comparison shows temperature-dependent trade-offs against independent fits, and the matched encoder comparison shows that exact spin-flip invariance can improve full TVD at some temperatures while worsening it at others. Supplementary runs establish the computational feasibility of classical optimization up to 2048 qubits. Residual full-distribution TVD and sampled MMD^2 errors limit claims of distributional accuracy. More expressive tractable encoders, adaptive kernels and evaluation of quantum deployment are directions for improving conditional generation; classical sampling hardness and practical quantum advantage remain open.

DATA AVAILABILITY

The code and data supporting this study are available on GitHub [48].

ACKNOWLEDGMENTS

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Appendix A: Hyperparameters and evaluation protocols

1. Common training settings

All trained models minimize the unbiased squared-MMD objective. Encoders remain fixed, and the circuit phases are initialized independently as $\theta_j \sim \mathcal{N}(0, s_\theta^2)$. Each update samples one fitting condition and one kernel bandwidth uniformly, then draws a target minibatch with replacement. Nonidentity masks are refreshed at every update, retaining the nonidentity-probability correction. Reference samples are independent of the target minibatch and shared across masks. Table IV summarizes the settings.

Each initialization resets the optimizer and circuit phases while retaining the target pools. A training seed includes all prescribed initializations; selection takes the lowest-scoring checkpoint across them. Reported uncertainty summarizes variation across seeds. Gaussian and scaling models use `Adam` with cosine learning-rate decay to one tenth of the initial rate. Curie–Weiss models use the optimizer selected by HPO with a constant learning rate. The coefficient b_1 is 0.9; $b_2 = 0.999$ and $\epsilon = 10^{-8}$ apply to the `Adam` variants. Weight decay is zero.

TABLE IV. Training settings for the numerical experiments. Curie–Weiss columns use full-distribution TVD for hyperparameter and checkpoint selection. Curie–Weiss optimizers, learning rates and initialization widths are selected from Table VI. Numerical values are rounded to three significant figures. The displayed expected mask weight 8 denotes 7.99. Updates and initializations refer to confirmation and evaluation; screening uses the smaller budget specified in Sec. A 3 a.

Setting	Gaussian	Shared product input	Shared spin-flip input	Independent conditional	Independent unconditional	Scaling
Updates per initialization	1,024	2,000	2,000	2,000	2,000	5,000
Initializations per seed	1	3	3	3	3	3
Evaluation seeds	5	5	5	5	5	3
Training targets per condition	512	1,024	1,024	1,024	1,024	10,000
Target minibatch	128	256	256	256	256	450
Training mask draws	1,024	2,048	2,048	2,048	2,048	512
Training reference draws	1,024	2,048	2,048	2,048	2,048	1,024
Expected mask weights	{1}	{2, 4, 8}	{2, 4, 8}	{2, 4, 8}	{2, 4, 8}	{2, 4}
Checkpoint interval	32	100	100	100	100	25
Optimizer	`Adam`	`Adam`	`Adam`	`AdamW ScheduleFree`	`Adam`	`Adam`
Learning rate	0.005	0.0310	0.0100	0.0603	0.00844	0.01
Learning-rate schedule	Cosine	Constant	Constant	Constant	Constant	Cosine
Initialization width s_θ	0.25	0.696	0.500	0.119	0.117	0.1
Coefficient b_1	0.9	0.9	0.9	0.9	0.9	0.9

For expected mask weight w, the Gaussian bandwidth is

$$\sigma(n, w) = \left[-2 \ln\left(1 - \frac{2w}{n}\right)\right]^{-1/2}, \qquad 0 < w < n/2. \tag{A1}$$

Here, n counts visible qubits, and the expectation precedes nonidentity conditioning. Bandwidths have equal weights in the objective. The Curie–Weiss weights are numerically $\{2, 4, 7.99\}$, displayed as $\{2, 4, 8\}$; all three bandwidths are strictly positive. Rotation angles are in radians. Full and profile TVDs use exact circuit probabilities at sixteen qubits; sampled-MMD diagnostics do not determine these errors.

2. Discrete Gaussian translation

The model has 16 visible qubits on an open 4×4 lattice, with 16 single-qubit and 24 pair phases. Its input is

$$|\phi_G(c)\rangle = \bigotimes_{i=1}^{16} R_Z(c - 1/2) R_Y(1.5) \, |0\rangle_i \, . \tag{A2}$$

The target profile is $\rho(k \mid c) \propto \exp[-(k - (5 + 6c))^2/8]$, with uniform probabilities within each Hamming-weight shell. Training uses $c \in \{0.0, 0.25, 0.50, 0.75, 1.0\}$ and 512 samples per condition, giving 2560 training samples per seed. We

select the checkpoint minimizing mean exact profile TVD at development conditions $c \in \{0.125, 0.375, 0.625, 0.875\}$, including initialization and checks every 32 updates through update 1024. Evaluation uses $c_j = (j + 1/3)/80$ for $j = 0, \ldots, 79$, with 1000 target draws per condition for auxiliary test-MMD diagnostics. Full and profile TVDs use exact probabilities; each aggregate error is the median across five independent seeds of the mean error over evaluation conditions. Figure 2 averages exact profiles over the five selected circuits at the development midpoints and summarizes their mean output Hamming weights over the separate evaluation grid.

3. Curie–Weiss models

The sixteen-qubit target is $p(x \mid \beta) \propto \exp[\beta(2|x| - 16)^2/32]$. All models use the complete bank of 16 one-body and 120 two-body phases, with no ancillas. The shared product model uses $R_Z(0.75\beta)R_Y(\pi/2)\,|0\rangle$ on every qubit. The shared invariant model uses its normalized +1 global spin-flip projection, as defined in Sec. IV B 1. Both fit at $\beta \in \{0.6, 0.8, 1.2, 1.4\}$, select on development temperatures $\{0.5, 0.7, 1.3, 1.5\}$, and report at $\{0.4, 1.0, 1.6\}$. Independent baseline families fit a separate circuit at each reporting temperature, using either the corresponding product input or $|0\rangle^{\otimes 16}$. Each HPO study selects one optimizer, learning rate and initialization width jointly for the three circuits in its baseline family.

Each fit uses 1024 training targets per fitting temperature and seed, minibatches of 256, and 2048 mask and reference draws per update. Auxiliary sampled-MMD diagnostics use 2048 validation targets per fitting temperature and 2048 independently drawn test targets per reporting temperature. Each diagnostic uses 2048 masks per condition and bandwidth and 2048 reference samples per condition shared across bandwidths. These diagnostics do not select models. Parameters and optimizer states use double precision; phase contractions use single precision. Table V gives the aggregate fitting resources.

TABLE V. Curie–Weiss fitting budgets per seed. Target counts denote training-pool entries; optimizer updates include all three initializations of each circuit during confirmation and evaluation. Each independent family fits three circuits, whereas each shared model fits one.

Model	Circuits	Training targets	Updates
Shared product input	1	4,096	6,000
Shared spin-flip input	1	4,096	6,000
Independent conditional	3	3,072	18,000
Independent unconditional	3	3,072	18,000

a. *Hyperparameter optimization*

We run four separate Optuna studies with a tree-structured Parzen estimator (TPE) [49], one for each model family. Each study has 60 trials and searches three hyperparameters: a categorical optimizer choice among `Adam`, `AdamWScheduleFree` and `SGDScheduleFree`, the constant learning rate, and the Gaussian initialization width [50, 51]. Table VI gives the search space. Equal trial counts compare the same number of proposed configurations; each independent-family trial fits three circuits per seed. As a random-search benchmark [52], 60 independent draws have probability $1 - 0.95^{60} \approx 0.954$ of sampling at least one configuration in the best-performing 5% of the chosen sampling distribution. This calculation motivates the budget rather than guaranteeing coverage for adaptive TPE proposals.

TABLE VI. Curie–Weiss HPO search space. Optimizer choice is categorical; learning rates and initialization widths are sampled logarithmically.

Hyperparameter	Options or range
Optimizer	`Adam`, `AdamWScheduleFree`, `SGDScheduleFree`
Learning rate, `Adam` variants	$[3 \times 10^{-4}, 10^{-1}]$
Learning rate, `SGDScheduleFree`	$[10^{-3}, 1]$
Initialization width s_θ	$[0.01, 0.7]$ radians

TPE uses 12 startup trials, including one reference configuration per optimizer, and batches of two proposals with sampler updates in proposal order. The reference learning rates are 0.01 for the `Adam` variants and 0.1 for `SGDScheduleFree`, all with $s_\theta = 0.5$. No trial is pruned. Optimizer coefficients are fixed as specified above, with no

warmup or gradient clipping. Schedule-free optimizers use learning-rate weighting exponent 2 and their evaluation parameters for checkpoint scoring.

Each screening trial uses three seeds, 1000 updates and one initialization per seed. Selection includes initialization and every 100 updates, using exact TVD over all 2^{16} bitstrings. A shared model selects the checkpoint minimizing mean development-temperature TVD. Each independent circuit selects at its fitted temperature; the three circuit errors are then averaged within each seed. The trial score is the median of the three seed scores, preserving mean aggregation across temperatures.

We confirm the three best distinct configurations on five fresh seeds, with 2000 updates and three initializations per seed. For each seed, we select the best checkpoint across its initializations, then choose the configuration with the lowest median score across the five seeds. Reporting reuses the five selected confirmation runs without further fitting or reselection. These seeds therefore do not form an independent final cohort. The current protocol excludes shared reporting temperatures from fitting and selection, although earlier experiments examined them; independent circuits fit at their reporting temperatures. Per-temperature results report means and sample standard deviations across the five seeds.

b. *Spin-flip invariant input and secondary diagnostics*

The two shared inputs use the same HPO protocol and paired seeds. Target pools, condition, minibatch, bandwidth and mask schedules, and Gaussian initialization directions are shared within each seed; initialization widths may differ after tuning. Reference sampling uses common random numbers with input-dependent laws. Odd masks remain in the objective even though their projected-model moments vanish.

For deterministic preparation of the invariant input, let $t = 0.75\beta$ and define $Z_\ell^{(p)} = [1 + (-1)^p \cos^\ell(t)]/2$, the parity-$p$ probability of ℓ independent reference bits. The last wire accumulates the parity p of the prepared prefix. With ℓ wires remaining, the next wire receives a parity-controlled rotation

$$\alpha_p = 2\,\mathrm{atan2}\left(\sin(t/2)\sqrt{Z_{\ell-1}^{(p\oplus 1)}}, \cos(t/2)\sqrt{Z_{\ell-1}^{(p)}}\right). \tag{A3}$$

A controlled-NOT (CNOT) gate updates the accumulator. Repeating on the first $n-1$ wires prepares the even-conditioned reference magnitudes. Applying a phase of $-\pi/2$ to each occupied wire gives Eq. (50); final Hadamards prepare Eq. (49). The construction uses $O(n)$ gates without ancillas or postselection.

The projected input produces

$$q_{\boldsymbol{\theta},+}(x \mid \beta) = \frac{|A_{\boldsymbol{\theta},\beta}(x) + A_{\boldsymbol{\theta},\beta}(\bar{x})|^2}{2[1 + \cos^n(t)]}, \tag{A4}$$

where $A_{\boldsymbol{\theta},\beta}(x) = \langle x|U_{\boldsymbol{\theta}}|\phi(\beta)\rangle$. Its interference term changes the model family, but does not guarantee improved fit. The invariant encoder alone gives $q_{0,+}(x \mid \beta) = [1 + \cos((2|x| - n)t)]/[2^n(1 + \cos^n(t))]$. Both encoder-only laws are uniform within Hamming shells, so their full and profile TVDs coincide.

TABLE VII. Exact Hamming-profile TVD for sixteen-qubit Curie–Weiss models. Trained entries give means ± sample standard deviations across 5 runs from the confirmation cohort used for model selection, reused for reporting; the encoder is deterministic. Shared reporting temperatures are excluded from fitting and current selection, while independent circuits fit at each reporting temperature. Bold entries mark the lowest mean error, including the encoder.

Model	Evaluation	Hamming-profile TVD ↓		
		$\beta = 0.4$	$\beta = 1.0$	$\beta = 1.6$
Encoder only	Baseline	0.1180	0.4406	0.8699
Shared conditional IQP	Held-out	0.0976 ± 0.0269	0.1301 ± 0.0317	$\mathbf{0.0631 \pm 0.0135}$
Independent conditional IQPs	Fitted	$\mathbf{0.0939 \pm 0.0073}$	0.0885 ± 0.0116	0.3948 ± 0.0003
Independent unconditional IQPs	Fitted	0.1186 ± 0.0040	$\mathbf{0.0780 \pm 0.0320}$	0.3293 ± 0.0119

Without the learned circuit, the product-state encoder has full-distribution TVDs of 0.1180, 0.4406, and 0.8699 at $\beta = 0.4, 1.0, 1.6$, respectively; the spin-flip invariant encoder gives 0.2561, 0.5536, and 0.9002. Input projection alone therefore does not improve the fit at these temperatures.

Table VII reports Hamming-profile errors. Table VIII reports paired full-distribution TVD differences between the learned inputs. We form each difference within a seed before computing the mean and sample standard deviation, so

TABLE VIII. Differences in exact full-distribution TVD between paired runs, spin-flip invariant input minus product-state input. Negative values favor the invariant input. Entries summarize 5 paired runs; SD denotes sample standard deviation. The confirmation cohort used for model selection is reused for reporting.

Statistic	$\beta = 0.4$	$\beta = 1.0$	$\beta = 1.6$
Mean	−0.1338	−0.0287	0.0096
Sample SD	0.0199	0.0154	0.0218

the spread differs from a difference of the separate standard deviations. The invariant input lowers full-distribution TVD in all five paired runs at $\beta = 0.4$ and 1.0, and in two of five at $\beta = 1.6$.

4. Scaling

The scaling models use the Curie–Weiss encoder on every wire, with no ancillas and open lattices listed in Table IX. At size n, the target law is $p(x \mid \beta) \propto \exp[\beta(2|x| - n)^2/(2n)]$. Training uses $\beta \in \{0.6, 0.8, 1.2, 1.4\}$, with 10,000 samples per condition and 40,000 per seed. Evaluation uses $\beta \in \{0.4, 0.95, 1.0, 1.05, 1.6\}$. We use three independent seeds at each size. Table IV gives the cosine-`Adam` settings, with 5,000 updates per initialization and three initializations per seed. These settings differ from the sixteen-qubit HPO winners; transfer or retuning remains untested.

TABLE IX. Scaling architectures, including single-qubit and nearest-neighbor pair phases.

Visible qubits	Lattice	Phases
64	8×8	176
128	8×16	360
256	16×16	736
512	16×32	1,488
1,024	32×32	3,008
2,048	32×64	6,048

Checkpoint selection minimizes mean sampled validation MMD^2 across training temperatures and bandwidths, checking initialization and every 25 updates across three initializations. Held-out temperatures do not enter fitting or selection. We report the selected initialization at its initial checkpoint and every 500 updates through update 5,000, averaging five independent MMD estimates per seed and condition grid. Each estimate uses 2,000 target draws per condition, 512 masks per condition and bandwidth, and 1,024 reference draws per condition. We reuse the reporting sample sets across iterations and share reference samples across bandwidths. The plot shows the median and interquartile range across three seed averages. We retain negative unbiased MMD^2 estimates.